



Path Energy Traffic Ratio API (PETRA)

draft-petra-green-api-02

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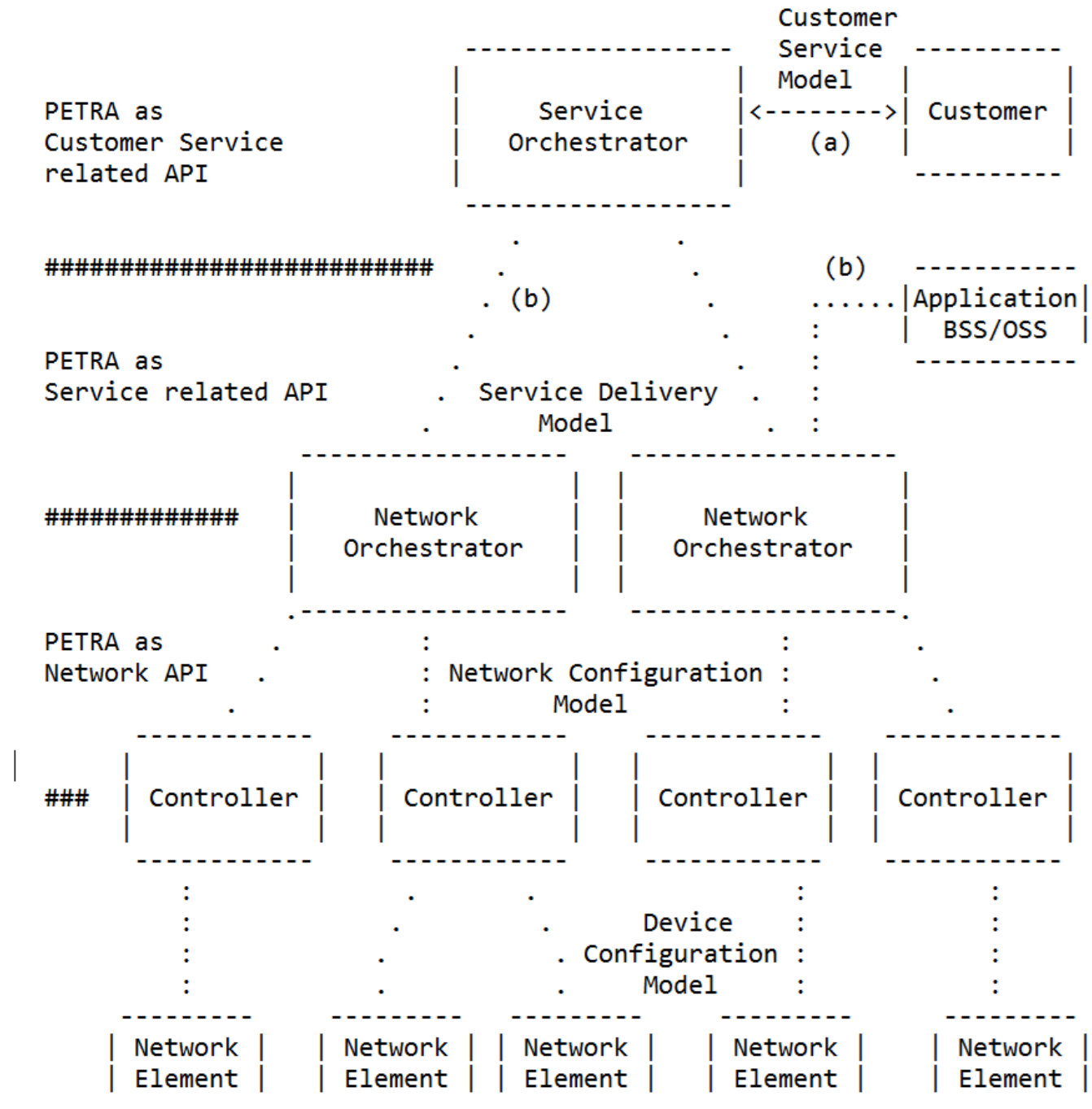
Motivation

- Provide visibility about energy consumption in a path
 - Metrics such as power consumption between source and destination (potentially related with throughput)
- Define an API that can provide such information
 - Using well-known architectures and schemas (e.g. YANG)
- This information can be consumed externally (e.g., SD-WAN customers) or internally (e.g., for operator optimization purposes)
- Draft presented in IETF 121, 123 and now 124
- Early version presented in PANRG at IETF 120
- Spin-off draft contributed to SUSTAIN RG at IETF 124

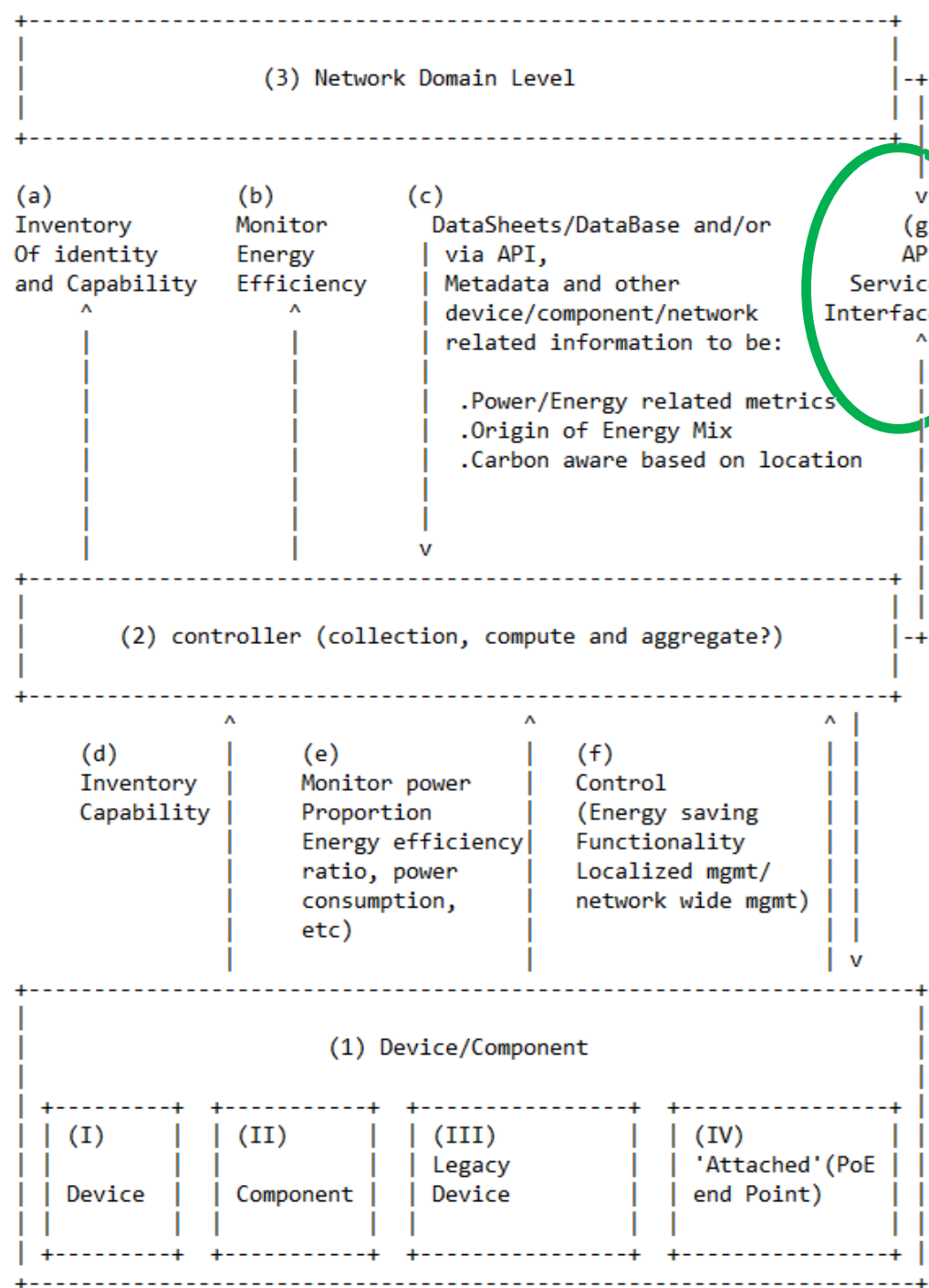
Changes in -02

- Refined wording
 - E.g., “energy management” instead of “sustainability”
- Consideration of end-points not represented by IP addresses
- Focus on Watts/Gb as the information retrieved by the API
- Illustrated the positioning of PETRA wrt RFC8309 (see next slide)
- Added security considerations
- New Annex (A.3) for “SLA Negotiation for Energy-Efficient Services”, and simplification of Annex B (figure now in framework document)
- Authors list updated to be aligned with the progress of the draft
 - Adrián now in front page
 - Alejandro and Fernando moved to contributors’ section, Roland also listed as contributor

Positioning of PETRA wrt RFC 8309



Green Framework + PETRA



PETRA as one of the possible APIs defined for this Interface of the framework

Next steps

- Gather and incorporate WG feedback
- Confirm validity of the approach
 - PETRA as one candidate API for the framework in order to retrieve energy-related information
- Pending
 - Explore incentive scenarios (expanding the initial insights on SLAs)
 - Generalize the model for being not only for IP end-points
 - Align with GREEN modelling efforts to ensure consistency

Backup

Path Energy Traffic Ratio API (PETRA)

- There can be multiple paths between origin and destination
- Energy consumption might be different on each path: dependent on device characteristics and architecture, transceiver bit rate, number of hops, etc.
- API Query:
 - <src-IP, dst-IP, throughput, ...>
- API Response:
 - <watts-per-gigabit, carbon-intensity, ...>

